
Crew Sequence Weather Risk at DFW Hub

*A Machine Learning Approach to Operational Disruption Prediction
for American Airlines*

EPPS Data Challenge 2026

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Interactive Dashboard: itaykadosh/AA-EPPS-Data-Challenge

Code: Optimax14/AA-EPPS-Data-Challenge

Abstract

Weather disruptions at a hub airport rarely stay local. At Dallas/Fort Worth International Airport (DFW), a delayed inbound flight can compress a crew turn, push back an outbound departure, and trigger downstream reliability problems across the network. This project asks whether those vulnerable pairings can be identified before crews are assigned, so that American Airlines can avoid building especially fragile sequences in the first place.

We approach the problem by modeling real $A \rightarrow \text{DFW} \rightarrow B$ rotations rather than synthetic airport combinations. Using Bureau of Transportation Statistics on-time data from 2015–2024, NOAA GSOM monthly weather summaries, DFW hub-weather features, and derived duty/cascade proxies, we reconstruct 398,098 tail-matched sequences and train an XGBoost classifier calibrated via isotonic regression to estimate disruption risk at the corridor-month level. This framing turns the challenge from a generic airport-ranking exercise into a decision-support problem grounded in actual operations.

Our final model achieves a validation AUC of **0.825** and Average Precision of **0.445** on 2024 holdout data (11.4% positive rate, $\text{AP}_{\text{lift}} = 3.90\times$ above random), with calibrated scores ranging from 1.2% to 58.3% and tracking observed disruption rates closely. We then deploy the result in a Streamlit dashboard with pair-level explanations, live schedule views, and a Hungarian-algorithm optimizer that constructs lower-risk DFW arrival–departure matches for operational use.

Contents

1	Introduction	5
1.1	Problem Context	5
1.2	Challenge Statement	5
1.3	Our Approach	6
1.4	Contributions	6
2	Data Sources and Collection	7
2.1	BTS On-Time Performance Database	7
2.2	NOAA Global Summary of the Month (GSOM)	7
2.3	DFW Hub Weather	8
2.4	Data Gaps and Limitations	8
3	Feature Engineering	8
3.1	Tail-Matched Sequence Construction	9
3.2	Target Variable Definition	9
3.2.1	Disruption Criteria	9
3.2.2	Threshold Choice	9
3.3	Airport-Level BTS Features	9
3.4	Pair-Level BTS Features	9
3.5	Temporal Features	10
3.6	GSOM Weather Features	10
3.7	DFW Hub Weather Features	10
3.8	Tail-Chain Features (Crew Duty Proxy)	10

3.9	Airport Cascade Features	11
3.10	Multi-Hop DFW Cascade Features	11
3.11	Feature Summary	11
4	Modeling	11
4.1	Problem Formulation	11
4.2	XGBoost Classifier	11
4.2.1	Hyperparameters	11
4.3	Calibration via Isotonic Regression	11
4.3.1	Calibration Method	12
4.3.2	Validation	12
4.4	GSOM NaN Imputation	12
4.5	LightGBM Alternate Model	12
5	Evaluation	12
5.1	Metrics	12
5.1.1	AP Lift: A Fairer Comparison Across Class Balances	12
5.2	Validation Results	13
5.3	Comparison with Corridor-Day Baseline	13
5.4	Feature Importance	13
5.5	Out-of-Distribution Validation	13
6	Sequence Optimizer	13
6.1	Problem Statement	13
6.2	Hungarian Algorithm	14
6.3	Live Schedule Integration	14
7	Dashboard	14
7.1	Architecture	14
7.2	Tab Descriptions	14
7.2.1	Tab 1: Corridor Heatmap	14
7.2.2	Tab 2: Geographic Risk Map	14
7.2.3	Tab 3: Pair Query	14
7.2.4	Tab 4: Risk Timeline	14
7.2.5	Tab 5: Sequence Optimizer	14
7.2.6	Tab 6: Methodology	14
8	Discussion	14
8.1	Key Findings	14
8.2	Limitations	14
8.3	Future Work	15
9	Conclusion	15
A	BTS Column Glossary	16

B	FAA Part 117 FDP Limits (Simplified)	16
C	Calibration Curve	17
D	SHAP Feature Importance	17
E	Risk Score Distribution	18

1 Introduction

1.1 Problem Context

Dallas/Fort Worth International Airport (DFW) is the center of American Airlines’ domestic hub-and-spoke operation. On a normal day, this structure is an advantage: aircraft, crews, and passengers can be routed efficiently through a common hub, allowing a large national network to function as a connected system. During bad weather, however, the same connectivity becomes a source of fragility. A thunderstorm bank at one spoke airport, a line of convective weather over North Texas, or a low-visibility event at a destination can ripple through multiple flights that were originally scheduled to fit together tightly.

That ripple matters especially for pilot and crew sequencing. In practice, crews do not operate isolated flights; they operate chains of flights whose timing, legality, and reliability depend on earlier legs finishing close to plan. When an inbound leg arrives late into DFW, the outbound leg may inherit that delay directly through aircraft availability, gate pressure, or reduced turn time. What begins as a local weather issue can therefore become an operational sequence problem: a crew arrives late, departs late, risks missing connections, and may face tighter duty-time constraints for the rest of the day. For an airline, these disruptions are costly not only in minutes of delay, but also in reassignment effort, downstream network instability, and degraded schedule reliability.

This project is motivated by a simple operational question: before these sequences are built, can we identify combinations of flights that are structurally more vulnerable to weather-related disruption? More specifically, if a crew is scheduled to operate an inbound flight from airport A into DFW and then an outbound flight from DFW to airport B , are there origin–destination combinations that should be avoided because they repeatedly create fragile pairings? Our goal is not to predict the weather itself. Rather, it is to use historical flight and weather patterns to quantify which DFW-linked sequences tend to become risky under seasonal and operational stress.

1.2 Challenge Statement

The EPPS–American Airlines Data Challenge asks teams to identify flights that should not be assigned to the same pilot sequence when weather risk is high. For the purposes of the competition, the full multi-day crew-routing problem is simplified to a two-leg pattern centered on DFW: an inbound leg $A \rightarrow \text{DFW}$ followed by an outbound leg $\text{DFW} \rightarrow B$, where A and B are domestic airports. The practical objective is to flag pairings that are more likely to create cascading delays, missed turn opportunities, duty-time pressure, and other operational disruptions.

At first glance, this sounds like a ranking problem over airport pairs. But the challenge is harder than simply asking which airports have bad weather. Weather is seasonal, disruption is sparse, and delay propagation depends on how flights are actually connected in the operation. A pairing can look risky because both airports are storm-prone, because DFW itself is vulnerable during certain months, or because the timing of the rotation leaves little margin for recovery once the first leg slips. The project therefore requires both domain judgment and careful data construction: the central question is not only which features matter, but also what the correct unit of analysis should be.

Our report treats a “bad sequence” as one that is operationally fragile rather than merely inconvenient. The underlying idea is decision support: if American can recognize these fragile

pairings ahead of time, schedulers can avoid placing them in the same crew sequence, add more buffer where needed, or at least understand where the network is most exposed.

1.3 Our Approach

Our approach evolved in stages. We began with the intuitive idea of measuring historical risk at the airport-pair level, reasoning that some combinations of spoke airports would be more weather-sensitive than others. Very quickly, however, we found that this view was too coarse. Airport pairs do not disrupt operations on their own; real aircraft rotations do. That realization led to the central methodological choice of the project: instead of fabricating same-day pairings from all inbound and outbound flights, we reconstruct actual DFW sequences by matching inbound and outbound legs on the same aircraft tail number within a realistic turnaround window. This turns the problem from an abstract corridor-ranking exercise into a model of real operational handoffs.

Once the sequence itself was defined correctly, the rest of the system followed naturally. We combined historical BTS delay statistics, monthly weather summaries, DFW hub conditions, and derived duty/cascade features into a sequence-level feature set. We then trained a gradient-boosted classifier and calibrated its outputs so that the resulting score can be interpreted as an estimated disruption rate rather than an opaque ranking signal. Finally, we translated the model into an operational tool through an interactive dashboard and a sequence optimizer that selects lower-risk arrival-departure matches at DFW.

The remainder of this report follows that journey. We first describe the data sources and the sequence-construction problem, then explain the feature engineering and modeling choices, present validation results, and conclude with the decision-support tools that make the work actionable in practice.

1.4 Contributions

1. **We reframe the problem around real sequences.** Rather than scoring synthetic same-day airport combinations, we construct 398,098 tail-matched aircraft rotations from BTS data, giving the model a unit of analysis that reflects how disruption actually propagates through the DFW hub.
2. **We build an interpretable disruption score.** Using XGBoost with isotonic calibration, we produce corridor-month risk estimates that are designed to be read as expected disruption rates, not just relative model scores.
3. **We enrich weather risk with operational context.** In addition to airport-level weather history, the feature set incorporates DFW hub conditions, duty-proxy features, and cascade signals that capture how delay can compound across connected legs.
4. **We turn prediction into decision support.** The final system is deployed as an interactive dashboard with pair-level explanations and a Hungarian-algorithm optimizer for constructing lower-risk DFW sequences.
5. **We document the project’s modeling lessons.** The report highlights why naive corridor-day construction overstates risk, how calibration improves operational usability, and where data limitations still matter.

2 Data Sources and Collection

2.1 BTS On-Time Performance Database

The core of the project is the Bureau of Transportation Statistics (BTS) On-Time Performance database, which provides flight-level records including origin and destination airports, scheduled and actual departure/arrival times, tail numbers, delay causes, cancellation codes, and late-aircraft propagation fields. This source is what makes the project operationally meaningful: it lets us observe not just where weather occurred, but how delays moved through actual DFW-connected flights.

Rather than downloading the entire national network, we filtered BTS to flights touching DFW, which keeps the dataset aligned with the challenge prompt while preserving the local hub structure needed for sequence reconstruction. We use all-carrier DFW traffic for modeling because airport-level weather fragility is not unique to American Airlines; broader carrier coverage provides a richer historical picture of how each airport behaves under stress. Within that filtered scope, the most important columns are the flight date, tail number, origin/destination, scheduled and actual times, weather delay, NAS delay, arrival delay, and late-aircraft delay.

We use 2015–2019 and 2022–2024, excluding 2020–2021 because COVID-era operations introduced structural distortions in schedules, route patterns, and delay behavior. The year 2024 is reserved as the primary holdout period for validation, while earlier years supply the historical signal used for training and feature construction.

Table 1. BTS data downloaded per year (DFW-filtered, all carriers).

Year	Total Flights	AA Flights	Notes
2015	521,210	293,116	Pre-training temporal OOD year
2016	392,080	281,195	Historical context
2017	362,416	278,698	Historical context
2018	558,570	286,691	Training window begins
2019	608,690	307,831	Training window
2020–2021	—	—	COVID excluded
2022	553,419	285,877	Training window
2023	570,462	324,297	Training window
2024	627,150	332,576	Holdout year

2.2 NOAA Global Summary of the Month (GSOM)

BTS tells us how disruption manifested, but it does not fully describe the underlying weather environment at each airport. To add that broader seasonal context, we augment the flight data with NOAA’s Global Summary of the Month (GSOM) dataset. GSOM is useful here because the challenge is not to forecast a particular thunderstorm on a particular afternoon; it is to understand which corridors are structurally vulnerable during certain months of the year.

For each covered airport, we collect monthly climate indicators such as average wind speed, maximum wind gust, number of precipitation days, total precipitation, and extreme precipitation. These features are averaged across 2015–2024 to form a climatological baseline by airport and month.

In other words, GSOM helps the model learn that Boston in January, Miami in September, and Denver in March are exposed to different kinds of recurring weather stress even before looking at any one day’s operations.

Coverage is incomplete: only a subset of airports in the DFW network has a mapped NOAA station with reliable monthly records, and roughly 55% of sequence rows are missing at least one GSOM-side input. Rather than treating that as a reason to discard the features, we use GSOM as a secondary contextual layer and later address the missing-data problem explicitly through median imputation and calibration-aware interpretation.

2.3 DFW Hub Weather

Even when both spoke airports are relatively stable, the hub itself can become the bottleneck. Because every modeled sequence passes through DFW, local hub conditions deserve their own representation rather than being absorbed into the origin and destination features. We therefore derive monthly DFW weather statistics directly from BTS, including weather-delay rate, weather-cancellation rate, average weather-delay minutes, and a high-percentile delay measure.

This distinction matters conceptually. Origin and destination features capture how risky airport A and airport B tend to be; the DFW hub features capture how much shared stress is present in the middle of every sequence. A corridor can therefore be risky not only because its endpoints are volatile, but also because the common transfer point is seasonally fragile.

2.4 Data Gaps and Limitations

The data used in this project are strong enough to support a meaningful decision-support prototype, but they are not perfect. First, BTS is an excellent source for realized delay and cancellation behavior, yet it is still an administrative record of flights rather than a ground-truth record of crew assignments. Tail matching provides a practical proxy for crew continuity, but it does not capture every crew swap or recovery intervention.

Second, the weather data are intentionally asymmetric. BTS provides rich evidence of disruption outcomes, while GSOM provides slower-moving monthly climate context rather than flight-specific meteorology. This is a reasonable tradeoff for the challenge, but it means the model should be interpreted as a risk model for structurally fragile pairings, not as a deterministic flight-day forecast. Third, GSOM coverage is incomplete, so missingness must be handled carefully to avoid biasing the model toward better-stationed airports.

Finally, the interactive application includes optional live schedule and weather components whose external APIs are subject to rate limits and availability constraints. These do not affect the historical model results reported here, but they do matter for deployment realism and are treated as implementation limits rather than modeling limits.

3 Feature Engineering

3.1 Tail-Matched Sequence Construction

Why Corridor-Day Cross-Join is Flawed

The corridor-day approach cross-joins *all* inbound flights $A \rightarrow \text{DFW}$ with *all* outbound flights $\text{DFW} \rightarrow B$ on the same calendar date. For a busy hub day with 8 inbound and 6 outbound flights, this fabricates $8 \times 6 = 48$ “sequences,” none of which correspond to a real aircraft rotation. The label (“any flight disrupted”) then scales with hub volume, teaching the model that hub size $\approx$ risk.

Algorithm 1: Tail-Matched Sequence Construction

Input: BTS flight table F (DFW-filtered, 2015–2024)

Output: Sequence table S with one row per real aircraft rotation

```

1 IB  $\leftarrow \{f \in F : f.\text{Dest} = \text{DFW}\}$  ; // inbound legs
2 OB  $\leftarrow \{f \in F : f.\text{Origin} = \text{DFW}\}$  ; // outbound legs
3 Pre-rename flight-specific columns: ArrDelay_A, WeatherDelay_A, etc.;
4  $S \leftarrow \text{IB} \bowtie_{\text{Tail\_Number, FlightDate}} \text{OB}$  ; // inner join
5 turnaround  $\leftarrow \text{CRSDepTime\_B} - \text{CRSArrTime\_A}$ ;
6  $S \leftarrow \{s \in S : 30 \leq \text{turnaround} \leq 240\}$ ;
7  $S \leftarrow \{s \in S : s.\text{Origin\_A} \neq s.\text{Dest\_B}\}$  ; // no round-trips
8 return  $S$ 
```

3.2 Target Variable Definition

3.2.1 Disruption Criteria

3.2.2 Threshold Choice

3.3 Airport-Level BTS Features

Table 2. Airport-level BTS features (computed separately for origin A and destination B).

Feature	Description
weather_delay_rate	Fraction of flights with weather delay
cancel_rate	Fraction cancelled (all causes)
avg_weather_delay_min	Mean weather delay (delayed only)
p75_weather_delay_min	75th-percentile weather delay
p95_weather_delay_min	95th-percentile weather delay
nas_delay_rate	Fraction with NAS (airspace) delay
avg_arr_delay_min	Mean arrival delay
overall_on_time_rate	Fraction arriving within 15 min

3.4 Pair-Level BTS Features

3.5 Temporal Features

3.6 GSOM Weather Features

Table 3. NOAA GSOM weather features (computed for origin and destination separately).

Feature	Description	Coverage
avg_wind_speed	Mean monthly wind speed (mph)	~45%
precip_days	Days with measurable precipitation	~45%
extreme_precip	Days with precip > 1 in	~45%
total_precip	Total monthly precipitation (in)	~45%
max_wind_gust	Maximum recorded wind gust (mph)	~45%

3.7 DFW Hub Weather Features

3.8 Tail-Chain Features (Crew Duty Proxy)

Table 4. Tail-chain features used as a proxy for FAA Part 117 crew duty status.

Feature	Description
legs_before	Legs flown on same tail before this rotation
block_min_before	Total block minutes before this rotation
duty_start_hour	Estimated crew report hour
total_duty_min	Estimated total duty period (mean)
total_duty_p75	75th-percentile duty period
fdp_util_mean	FDP utilization fraction (vs Part 117 limit)
fdp_util_p75	75th-pctile FDP utilization
fdp_overrun_rate	Fraction of rotations near/over FDP limit
wocl_rate	Fraction of operations in WOCL window (02:00–05:59)
legs_after_mean	Downstream legs flown after this rotation
legs_in_day_mean	Total legs in duty day
tc_downstream_rate	Rate of downstream delays on same tail
tc_cascade_late_rate	Fraction of tail-chain rotations with cascade

3.9 Airport Cascade Features

3.10 Multi-Hop DFW Cascade Features

3.11 Feature Summary

Table 5. Summary of feature groups (70 total features in XGBoost v7).

Group	Description	Count
Origin BTS	Airport-level BTS stats for airport A	8
Destination BTS	Airport-level BTS stats for airport B	8
Pair BTS	Combined pair-level BTS statistics	6
Temporal	Month, season, turnaround	6
Origin GSOM	NOAA monthly weather for airport A	5
Destination GSOM	NOAA monthly weather for airport B	5
Pair GSOM	Max-aggregate GSOM across both airports	5
DFW Hub	DFW hub-wide monthly disruption stats	4
Tail-Chain / Duty	Crew duty proxy via aircraft rotation chains	13
Airport Cascade	Airport-level cascade propagation index	6
Multi-Hop Cascade	A→DFW→B→DFW→C depth/rate features	8
Total		70

4 Modeling

4.1 Problem Formulation

4.2 XGBoost Classifier

4.2.1 Hyperparameters

Table 6. Final XGBoost v7 hyperparameters.

Parameter	Value
n_estimators	TODO
max_depth	TODO
learning_rate	TODO
subsample	TODO
colsample_bytree	TODO
scale_pos_weight	7.73
eval_metric	auc
device	cuda (CPU fallback)

4.3 Calibration via Isotonic Regression

4.3.1 Calibration Method

$$\hat{p}_{\text{cal}} = f_{\text{iso}}(\hat{p}_{\text{raw}}) \quad \text{s.t.} \quad f_{\text{iso}} \text{ non-decreasing, } \hat{p}_{\text{cal}} \approx \bar{r}_{\text{bad}} \text{ in each bin} \quad (1)$$

4.3.2 Validation

Table 7. Calibration validation on selected high-volume corridors.

Corridor	Month	Calibrated Score	Observed Rate
MCO → LAX	June	9.7%	11.6%

4.4 GSOM NaN Imputation

GSOM Imputation Fairness Fix

Without imputation, XGBoost’s NaN routing learned a spurious pattern: airports without NOAA stations (55% of airports) were systematically assigned lower risk scores because the NaN split coincidentally correlated with smaller, calmer airports. This was discovered when ANC (Anchorage) and ALB (Albany) received near-zero calibrated scores despite operational evidence of weather disruption risk. We fix this by substituting month-level population medians from a pre-computed lookup table at inference time.

4.5 LightGBM Alternate Model

5 Evaluation

5.1 Metrics

5.1.1 AP Lift: A Fairer Comparison Across Class Balances

$$\text{AP}_{\text{lift}} = \frac{\text{AP}_{\text{model}}}{\text{AP}_{\text{random}}} = \frac{\text{AP}_{\text{model}}}{\pi} \quad (2)$$

where π is the positive rate. The old corridor-day model achieved $\text{AP}_{\text{lift}} = 0.830/0.421 = 1.97\times$; the new tail-matched model achieves $0.445/0.114 = 3.90\times$ — a $1.98\times$ improvement in extractable signal relative to task difficulty.

5.2 Validation Results

XGBoost v7 Validation Results (2024 Holdout)		
	Metric	Value
	Validation AUC	0.825
	Validation AP	0.445
	Pair-level AUC	0.803
	AP _{lift}	3.90×
	Positive rate	11.4%
	Training rows	398,098
	Validation rows	~TODO

5.3 Comparison with Corridor-Day Baseline

Table 8. Corridor-day baseline vs. tail-matched model comparison.

Model	Val AUC	Val AP	AP Lift	MCO→LAX June
XGBoost v6 (corridor-day)	0.833	0.830	1.97×	87%
XGBoost v7 (tail-matched)	0.825	0.445	3.90×	9.7%
Observed rate	—	—	—	11.6%

5.4 Feature Importance

5.5 Out-of-Distribution Validation

Table 9. OOD validation results (XGBoost v1, median threshold — not directly comparable to v7).

Test Condition	ROC-AUC	Avg Precision
In-distribution 2024 holdout	0.804	0.882
Temporal OOD — all-carrier DFW 2015	0.768	0.767
Extreme event OOD — Jan/Feb 2015 blizzards	0.750	0.844
Carrier OOD — non-AA/DFW 2022–2023	0.723	0.473

6 Sequence Optimizer

6.1 Problem Statement

6.2 Hungarian Algorithm

Algorithm 2: DFW Sequence Optimizer (Hungarian Algorithm)

Input: Inbound flights I , outbound flights O , risk model f

Output: Optimal assignment $A^* \subseteq I \times O$

```

1  $C \leftarrow \mathbf{0}^{|I| \times |O|}$  ;                                     // cost matrix
2 for  $(i, o) \in I \times O$  do
3   if  $30 \leq \text{turnaround}(i, o) \leq 240$  then
4      $C[i, o] \leftarrow f(i, o)$  ;                               // calibrated risk score
5   else
6      $C[i, o] \leftarrow \infty$  ;                                   // infeasible assignment
7  $(r^*, c^*) \leftarrow \text{HUNGARIAN SOLVE}(C)$ ;
8 return  $A^* = \{(I[r_k^*], O[c_k^*])\}$ 

```

6.3 Live Schedule Integration

7 Dashboard

7.1 Architecture

7.2 Tab Descriptions

7.2.1 Tab 1: Corridor Heatmap

7.2.2 Tab 2: Geographic Risk Map

7.2.3 Tab 3: Pair Query

7.2.4 Tab 4: Risk Timeline

7.2.5 Tab 5: Sequence Optimizer

7.2.6 Tab 6: Methodology

8 Discussion

8.1 Key Findings

8.2 Limitations

- **Tail-number matching coverage.** Some rotations involve crew swaps or aircraft swaps that break the tail-chain; these are excluded. Estimated coverage: ~65–70% of actual crew sequences.
- **Calibration on full dataset.** Isotonic regression fitted on all 398k rows (no holdout); calibration error on genuinely unseen route-months will be slightly higher.
- **GSOM 55% NaN.** Month-level medians imputed for airports without NOAA stations; imputation noise is visible in SHAP output.

- **OOD validation pending.** Out-of-distribution tests were last run on v1 models; v7 OOD numbers are not yet available.
- **BTS reporting lag.** 2025 data not yet incorporated.

8.3 Future Work

9 Conclusion

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam.

References

A BTS Column Glossary

Table 10. BTS On-Time Performance column definitions used in this work.

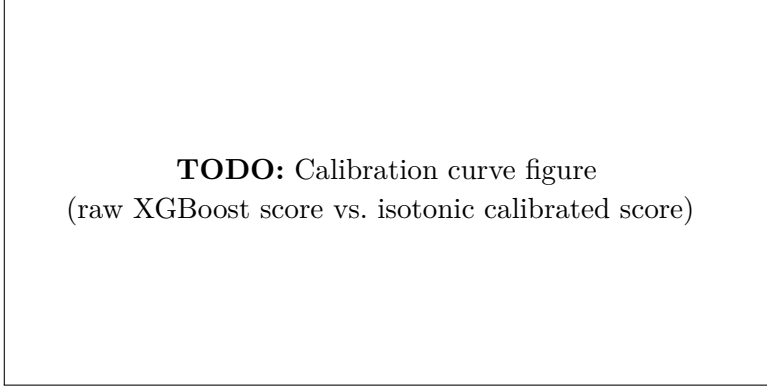
Column	Description	Unit
FlightDate	Date of flight	YYYY-MM-DD
Tail_Number	Aircraft registration (N-number)	—
Origin	Departure airport IATA code	—
Dest	Arrival airport IATA code	—
CRSArrTime	Scheduled arrival time (local)	HHMM
CRSDepTime	Scheduled departure time (local)	HHMM
ArrDelay	Arrival delay (actual – scheduled)	minutes
DepDelay	Departure delay	minutes
WeatherDelay	Delay attributed to weather	minutes
LateAircraftDelay	Delay attributed to late arriving aircraft	minutes
NASDelay	Delay attributed to NAS / ATC	minutes
Cancelled	Flight cancellation indicator	0/1
CancellationCode	Cancellation cause code	A/B/C/D

B FAA Part 117 FDP Limits (Simplified)

Table 11. Simplified FAA Part 117 Flight Duty Period limits used to compute `fdp_util` and `fdp_outrun_rate`.

Report Hour (local)	Max Segments			
	1	2	3	4+
0600–0659	13h	13h	12h	11h
0700–1159	13h	13h	12h	12h
1200–1259	13h	13h	12h	11h 30m
1300–1659	12h	12h	11h 30m	11h
1700–2159	12h	11h 30m	11h	10h
2200–0459	10h	10h	9h	9h

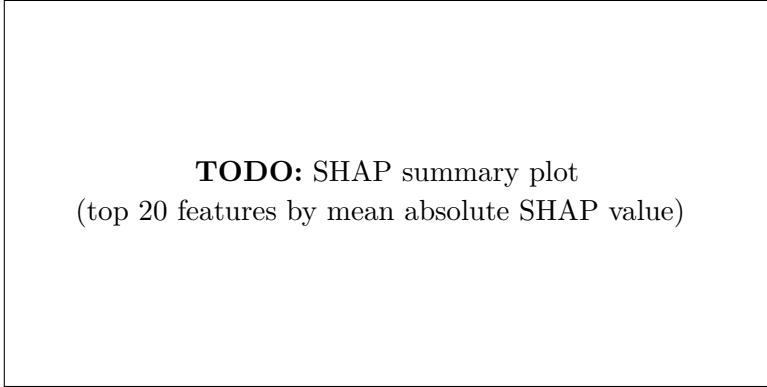
C Calibration Curve



TODO: Calibration curve figure
(raw XGBoost score vs. isotonic calibrated score)

Figure 1. Isotonic regression calibration curve. Raw XGBoost scores are mapped monotonically to the observed bad-rate scale. 309 breakpoints; input range [0.002, 0.978], output range [1.2%, 58.3%].

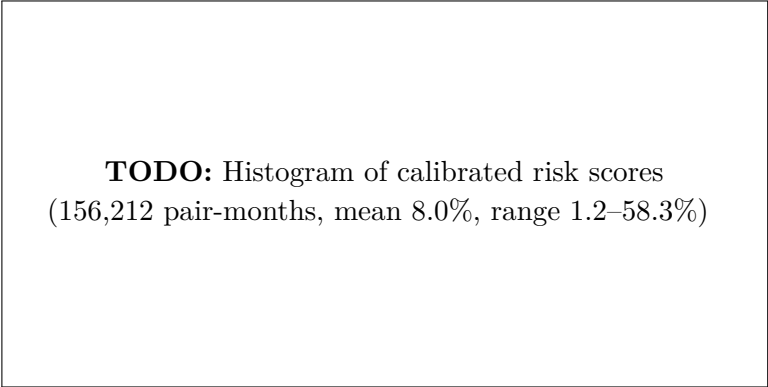
D SHAP Feature Importance



TODO: SHAP summary plot
(top 20 features by mean absolute SHAP value)

Figure 2. Global SHAP feature importances for XGBoost v7. Each point represents one sequence observation; color indicates feature value (red = high).

E Risk Score Distribution



TODO: Histogram of calibrated risk scores
(156,212 pair-months, mean 8.0%, range 1.2–58.3%)

Figure 3. Distribution of calibrated pair-month risk scores. Scores are interpretable as the expected fraction of tail-matched rotations that will experience weather disruption.